

Ore Pass

03/10/2019

Principle I

- *Ore pass is an opening that are driven in rock mass for conveyance of ore, waste rock, fill material from upper level or stope to lower level with help of gravity.
- *Ore pass can be vertical or incline (up to 50 degree).
- *Gravity assists flow of material.
- *Ore pass is an integrated part of material handling system in underground mining.
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Principle II

- *Ore pass can be excavated by raising or winzing.
- *Ore passes can be rectangular, how ever squire and circular ore passes (raise borer) are also available.
- *Borer driven circular ore pass is more stable than a squire or rectangular ore pass.

Principle II

*Empirical Relation for ore.

$L = 4.6\sqrt{(d^2k)}$ L= Side length for sq section.

$D = 5.2\sqrt{(d^2k)}$ D= Diameter for circular section.

d=size of largest lump,

k= 0.6 when content of sticky fines = 0%

k= 1 when content of sticky fines = 5%

k= 1.4 when content of sticky fines = 10%

In hard rock mines all very fine material (passing 200 mesh) is considered sticky due to water sprayed on it to suppress dust.

Ore Pass Excavation I

- *Ore pass are excavated by Raise Borer, Alimak raise climber, drop raising (long hole drilling), Manual raising.
- *Alimak raise climber allows support system to be established in ore pass.
- *Alimak raise climber does not need upper level to be available for ore pass excavation.
- *Raise borer does not disturb ground, while excavating raise.

Ore Pass Excavation II

- *Ore pass excavated by drop raising are shorter in length in compression to raise borer & Akimak Raise climber.
- *Ore pass shorter than 100m are excavated by long hole blasting.
- *Length of ore pass also depends on past experience of mine operator.
- *An excavation of long ore ore pass likely to intersect of poor ground.

Ore Pass Excavation III

- *Longer section of ore pass results high flow velocity of ore & compaction.
- *It is preferable to have vertical ore pass less than 100m.
- *To have better life of ore pass, it is preferable to excavate ore pass in relatively stable ground.
- *Structural failure of wall by material flow.
- *Ore pass wall wear due to abrasion.
- *Ore Pass damage because of hang up clearance by blasting.
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Ore Pass Excavation IV

- *Ore pass deteriorate faster, when density of ore is higher.
- *Ore pass excavated in higher stress area shows faster deterioration & it is better to excavate two ore pass than one.
- *When ore is very hard & ore pass is excavated in relatively softer ground, it is better to have two ore pass.

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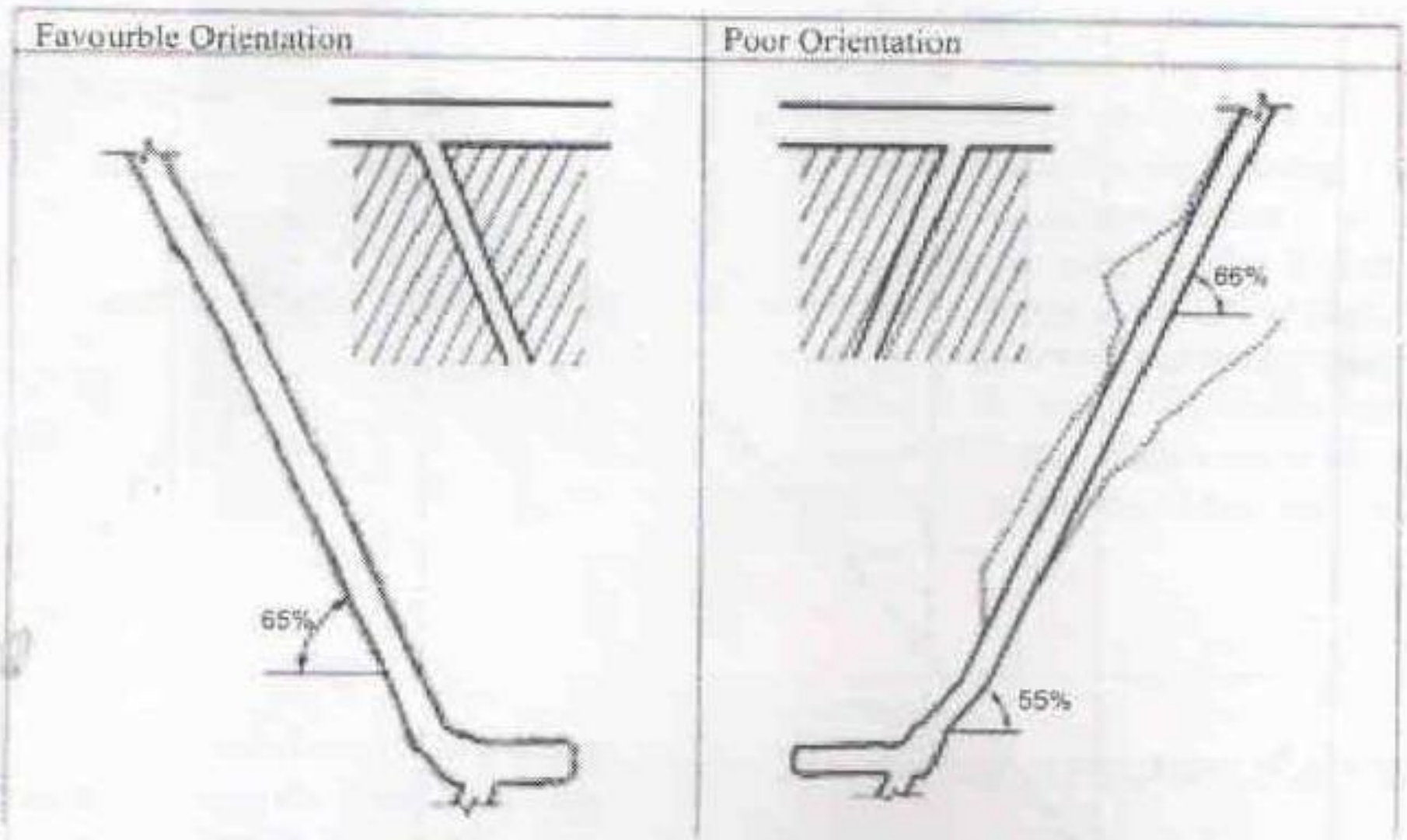
Ore Pass Excavation V

- *It is important to excavate ore pass perpendicular to bedding plane as much as possible.
- *Ore pass along bedding plane deteriorates faster.
- *Ore pass wall damage is mostly done by impact loading specially at intersection of finger raise.
- *Presence of structural defect in rock mass influences deterioration due to impact loading.
- *Ore pass with few fracture in wall does not endanger structural integrity of ore pass.

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Ore pass In Bedded Deposit I



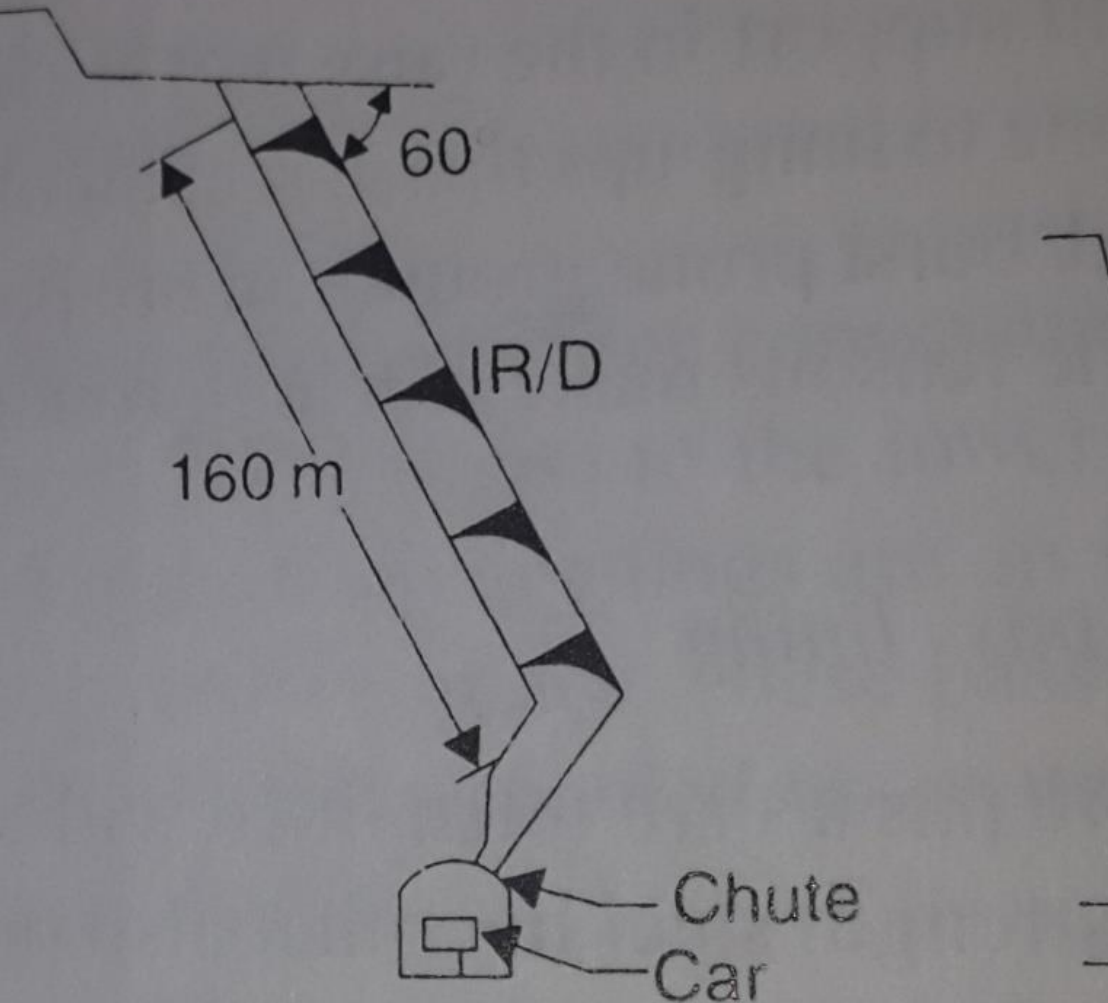
ore pass development with respect to bedding

Incline Ore Pass

- *Inspection raise and drifts are placed along ore pass for judging condition in ore pass. This provision also helps to remove jam / repairing work in ore pass.
- *Normally life of incline ore pass is longer because velocity of ore is low.

Incline Ore Pass

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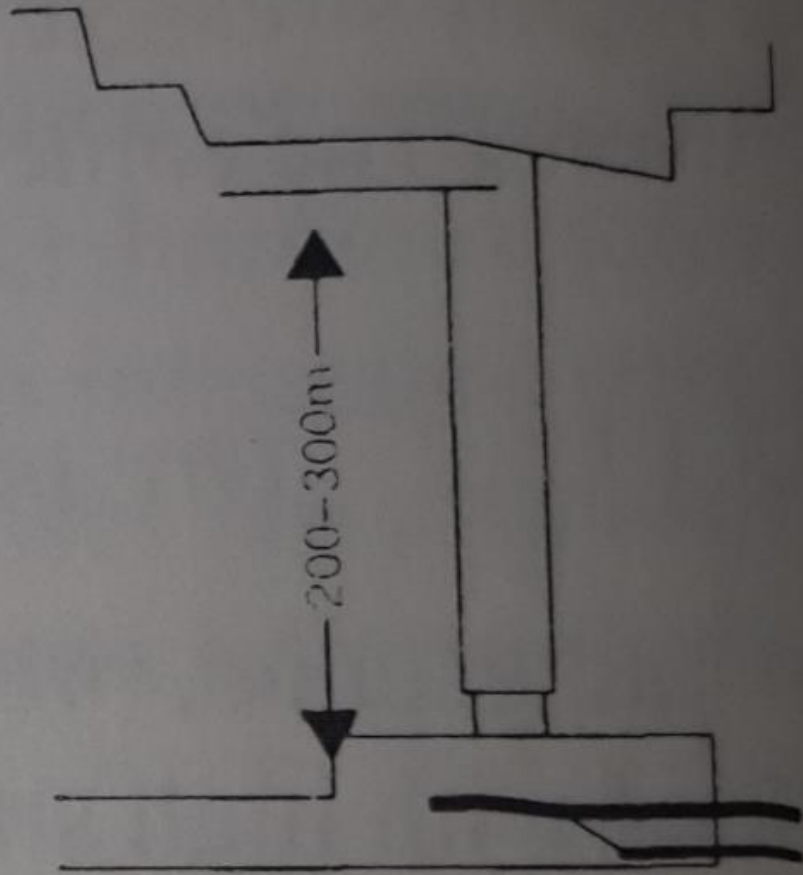
(e) Inclined orepass with inspection raises and drifts

Vertical Ore Pass

- *Vertical ore pass is long and mostly associated with shaft hoisting system.
- *To have better stability, these ore passes are made circular.
- *At bottom of ore pass, chute or gate are provided for controlling ore flow.

Vertical Ore Pass

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(f) Vertical circular
orepass

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Shape I

- *Normally ore pass is square or rectangular in cross section.
- *Circular section ore pass is achieved by raise borers.
- *Alimak raise climber can also excavate circular ore pass.
- *Experience of miners with existing ore pass is main factor for choosing among square or circular section.

Shape II

- *In case of high stress, shape or ore pass does not retain at circular or rectangular shape with flow of material.
- *Ore pass normally of 4 to 6 sqm in cross section.
- *Cross section of ore pass & change in cross section along with size of material influence hang up in ore pass.
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Finger Raise I

- *Finger raises are used to funnel material in to ore pass, intersecting, two or more production levels.
- *In case of ore pass serving one production level / point (stope), ore is dumped right at top opening of ore pass / chute.
- *Ore pass with single dumping point are of short length.
- *Generally finger raises are square section, with a smaller cross section area, than ore pass it self.

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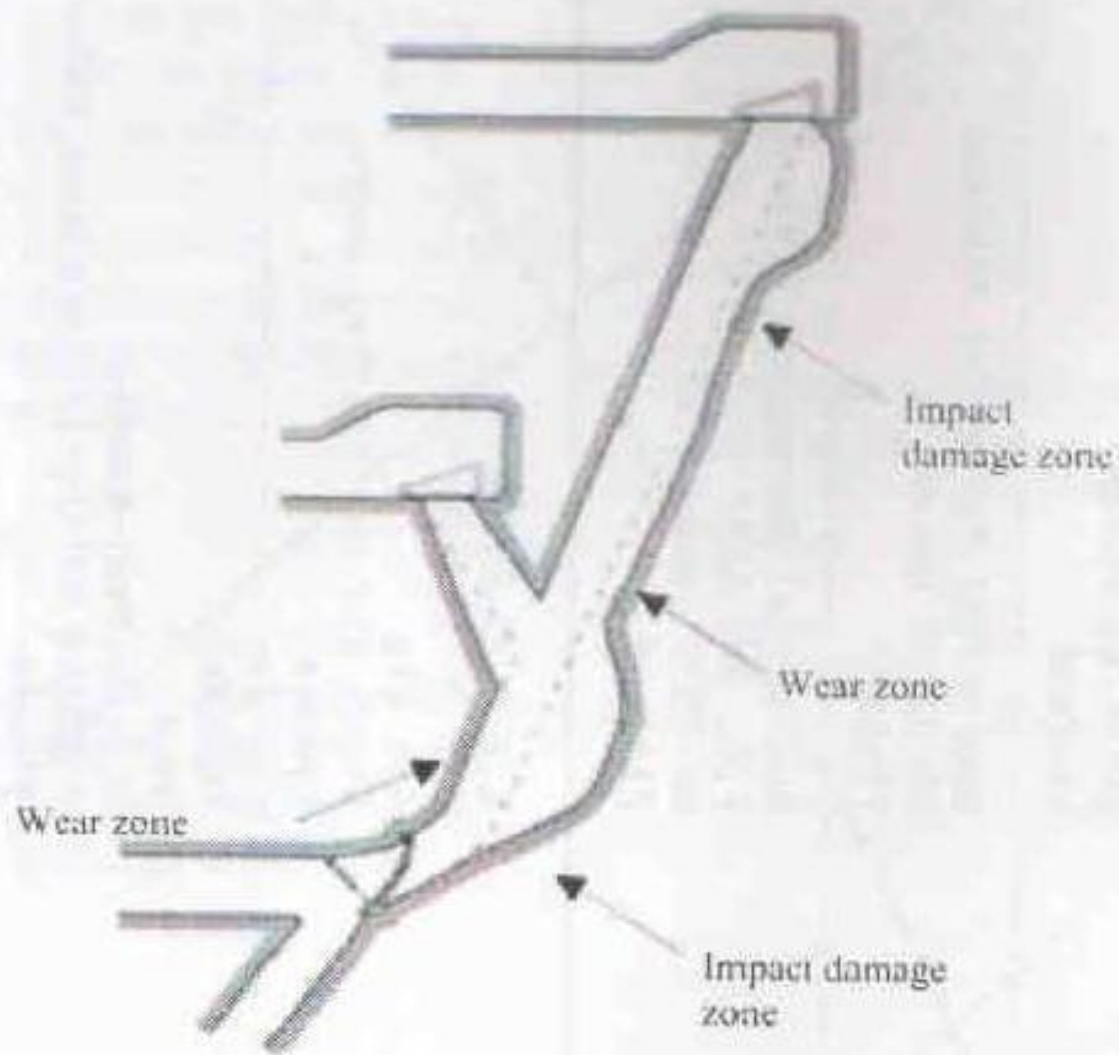
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Finger Raise II

- *It is quite unusual to have more than 3 finger raises for an ore pass.
- *Finger raises often results localized impact zone causing damage to ore pass wall & integrity of ore pass.
- *Damage to ore pass wall will get farther aggravated if excavation of finger raise is done using long hole blasting due to high impact.
- *Slop of finger raise at junction point with ore pass should be as gentle as possible, keeping flow of ore to minimize damage to ore pass wall.

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Damage By Finger Raise I



-Damage zones in an ore pass

Grizzly I

- *Over size material dumped into ore pass may lead to blockages or interlocking hang up.
- *This is avoided by installing necessary infrastructure to stop entry of over size material in ore pass.
- *On top of ore pass / mouth of finger raise grizzly is constructed to arrest over size material from going in.
- *Large size boulder in ore pass damages ore pass fast.
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Parallel Grizzly II

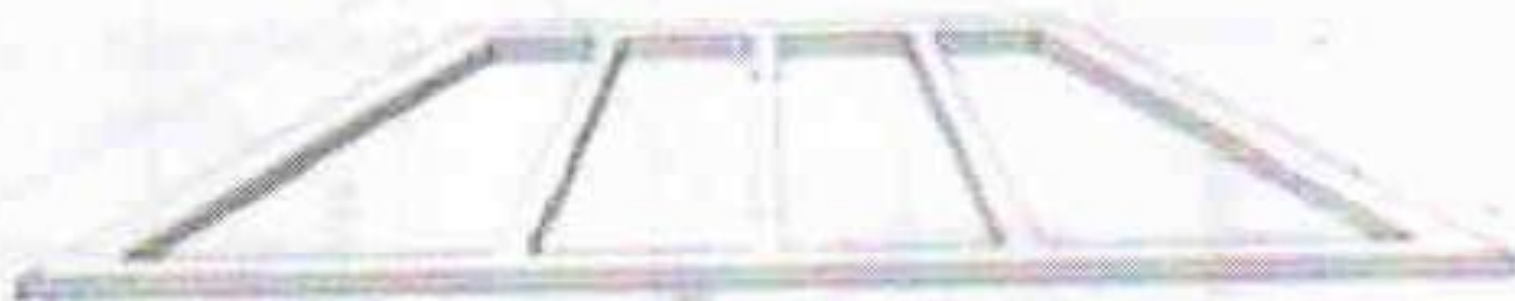
- *Parallel square section bar / rail are fixed at mouth of ore pass to arrest over size at mouth of ore pass.
- *Bars are fixed in line of direction of dumping & slopped slightly (5 to 7 degree) in direction of dumping.
- *This slope helps to use larger part of grizzly.
- *Steel bars are provided with protective wear plate, so that bar is protected from wear.

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Grizzly Parallel Bar I

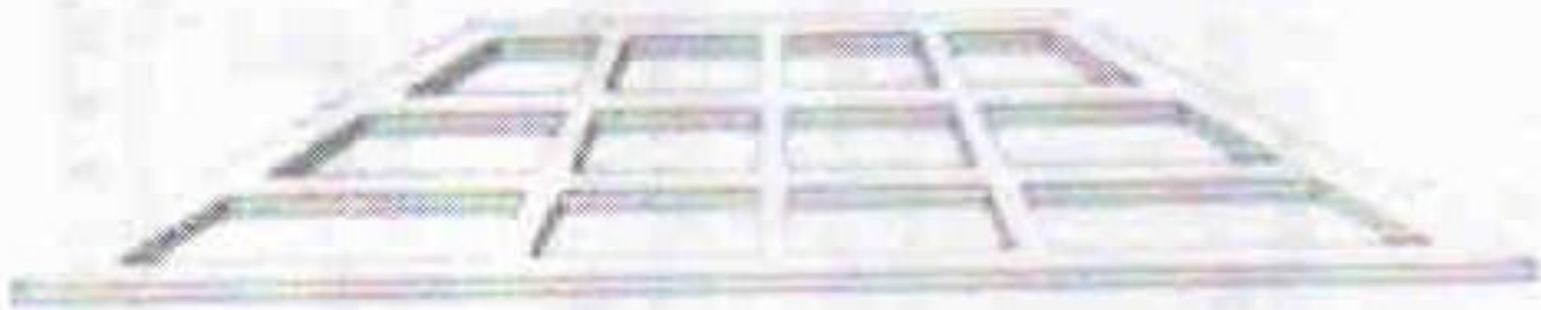


a) Scalper

Squire Grizzly III

- *Squire grizzly is used in place of parallel bar, where size restriction has to be closely maintained.
- *Squire grizzly requires less maintenance in compared to parallel bar grizzly.
- *Squire pattern provides more rigidity & stability to grizzly.
- *In squire grizzly, some time plugging takes place, making ore flow difficult.
- *Squire grizzly some time requires more racking.
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Grizzly Box I



b) Grizzly

Circular Grizzly IV

- *There are grizzly with circular opening (mantels).
- *Flow of ore is better in circular grizzly.

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c) Mantle

Grizzly V

- *Bigger boulder stuck on grizzly are hammered to smaller size.
- *If boulder is very big, then it has to be blasted over grizzly.
- *Blasting of boulder over grizzly should be avoided as much as possible, blasting damages grizzly.
- *Gap in between bar / rail should be 40% less than size of boulder, that can be handled at end of ore pass.
- *In case, ore pass has to handle large volume of ore through grizzly, hydraulic rock breaker has to be installed at grizzly.

Grizzly VI

- *Normally, grizzly men try to turn boulder, so that it can pass through without breaking.
- *Care should be taken to repair damaged grizzly as soon as possible.
- *Fallen broken bar or piece of grizzly, in ore pass creates obstruction in ore flow especially at discharge end.
- *Water spray & dust trap / collector is used at ore pass mouth (grizzly) to suppress dust.

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Grizzly VII

- *A barrier is erected near grizzly, so that LHD, LPDT does not fall in to grizzly / ore pass.
- *In case of mine car dumping, a lifting gear (chain pulley, crane) are placed at grizzly to lift derailed car, if any.
- *High pressure water jet is used to dislodge any jammed ore in LPDT, LHD, mine car at time of discharging ore to ore pass.
- *Water consumption in grizzly should be controlled.
- *Excess water in grizzly creates problem in ore hoisting.

Grizzly VIII

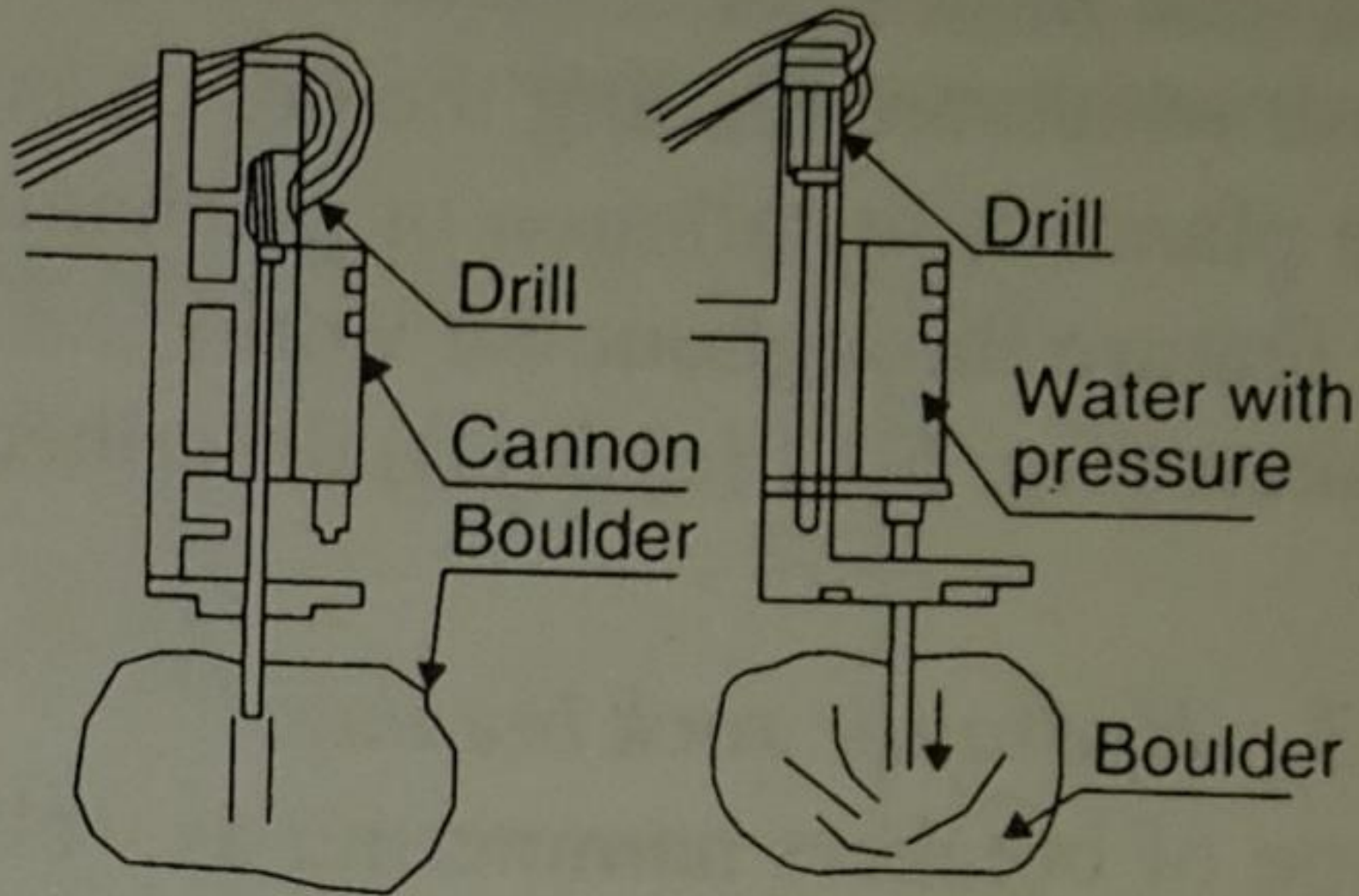
- *Persons working at grizzly must use safety belt to prevent accidental fall in ore pass.

- *Grizzly area should be lighted adequately.

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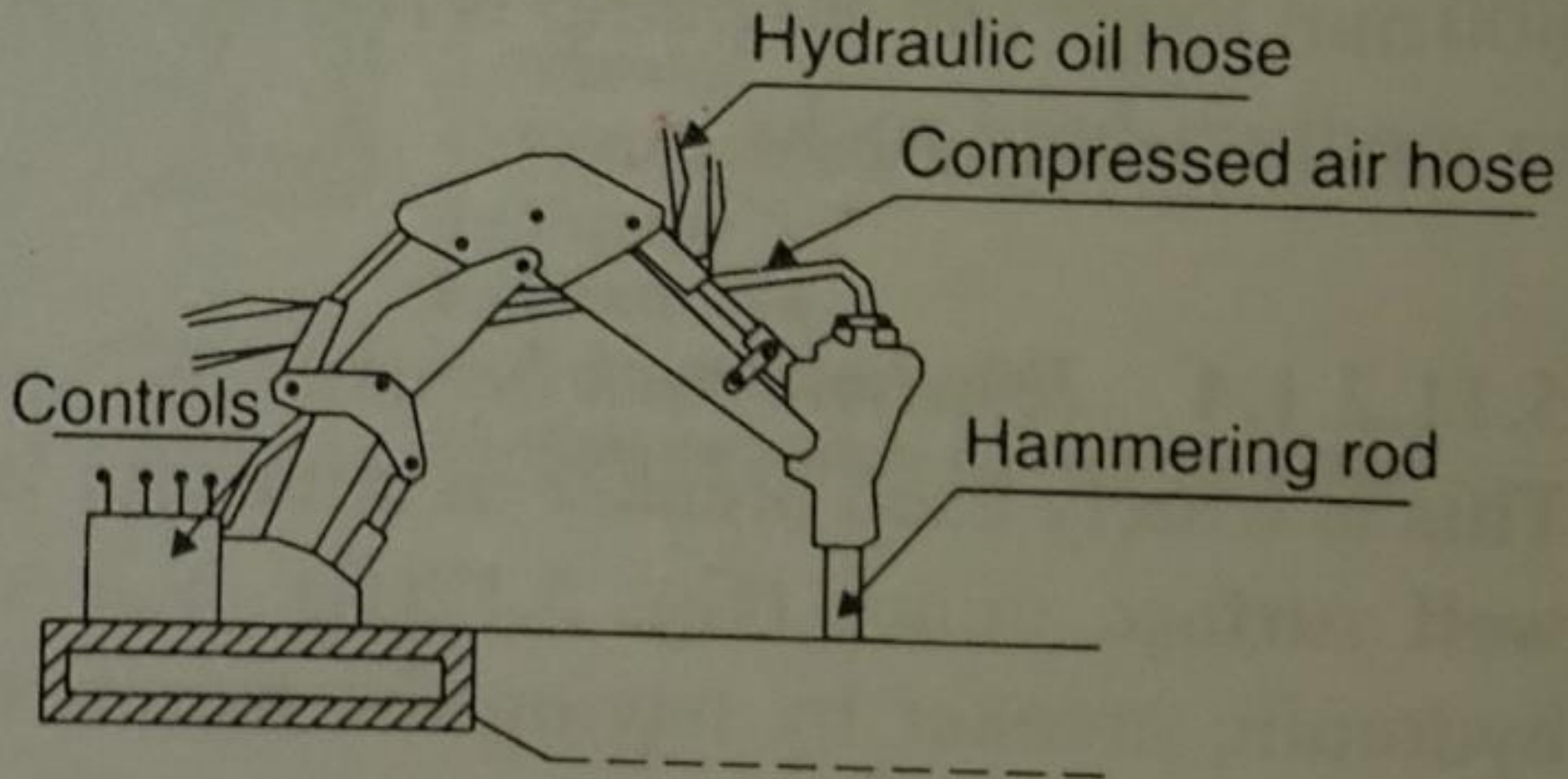
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Hydraulic (Water) Boulder Breaking



(d) Rock breaking hydraulically

Rock Breaker Boulder Breaking



- (f) Teledyne (mechanical) rock breaker hammering pneumatically but booms' movement is hydraulically.

Flow Of Ore I

- *Heavy weight circular chain (control chain) are suspended from head block.
- *Weight & number of chain regulate flow of ore from ore pass.
- *Ore flow is controlled by changing number of chain.
- *Number of chain is changed by lifting & keeping aside chain.
- *Chute with gate hinged at top physically stops movement of ore, when pushed by hydraulic cylinder.

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Flow Of Ore II

*Some time big boulder gets stuck at chute gate, keeping gate open, hense fail to control flow effectively.

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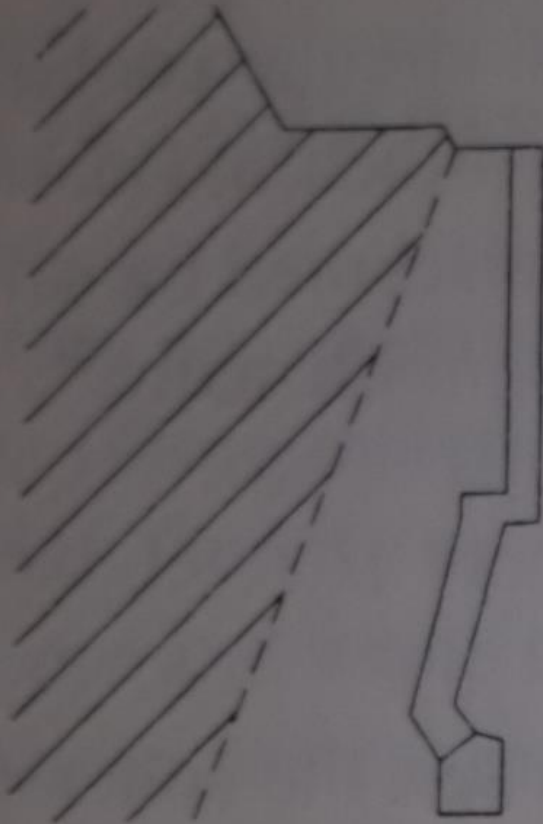
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Flow Of Ore III

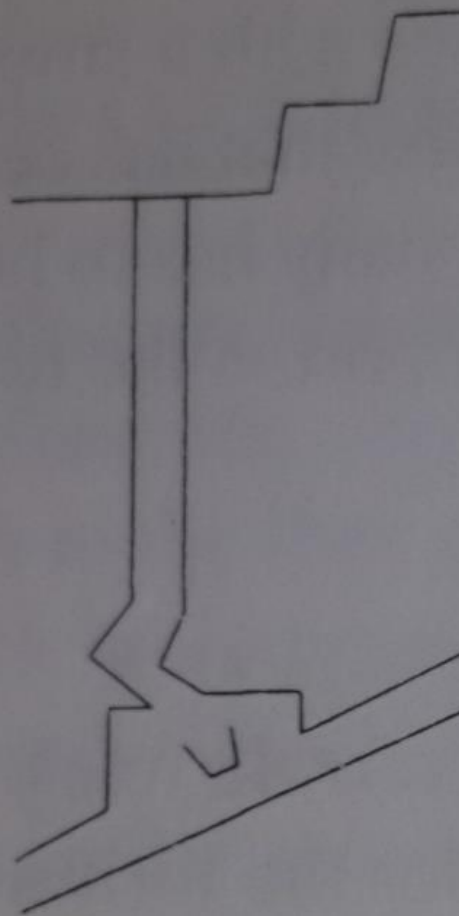
- *Ore pass with knuckle for controlling flow of ore in ore pass.
- *Some time ore pass gets jammed at knuckle.
- *Access is created up to knuckle with help of raise to clear any jam at knuckle.

Flow Control I

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(c) Orepass with
midway knuckle



(d) Orepass with
knuckle at bottom

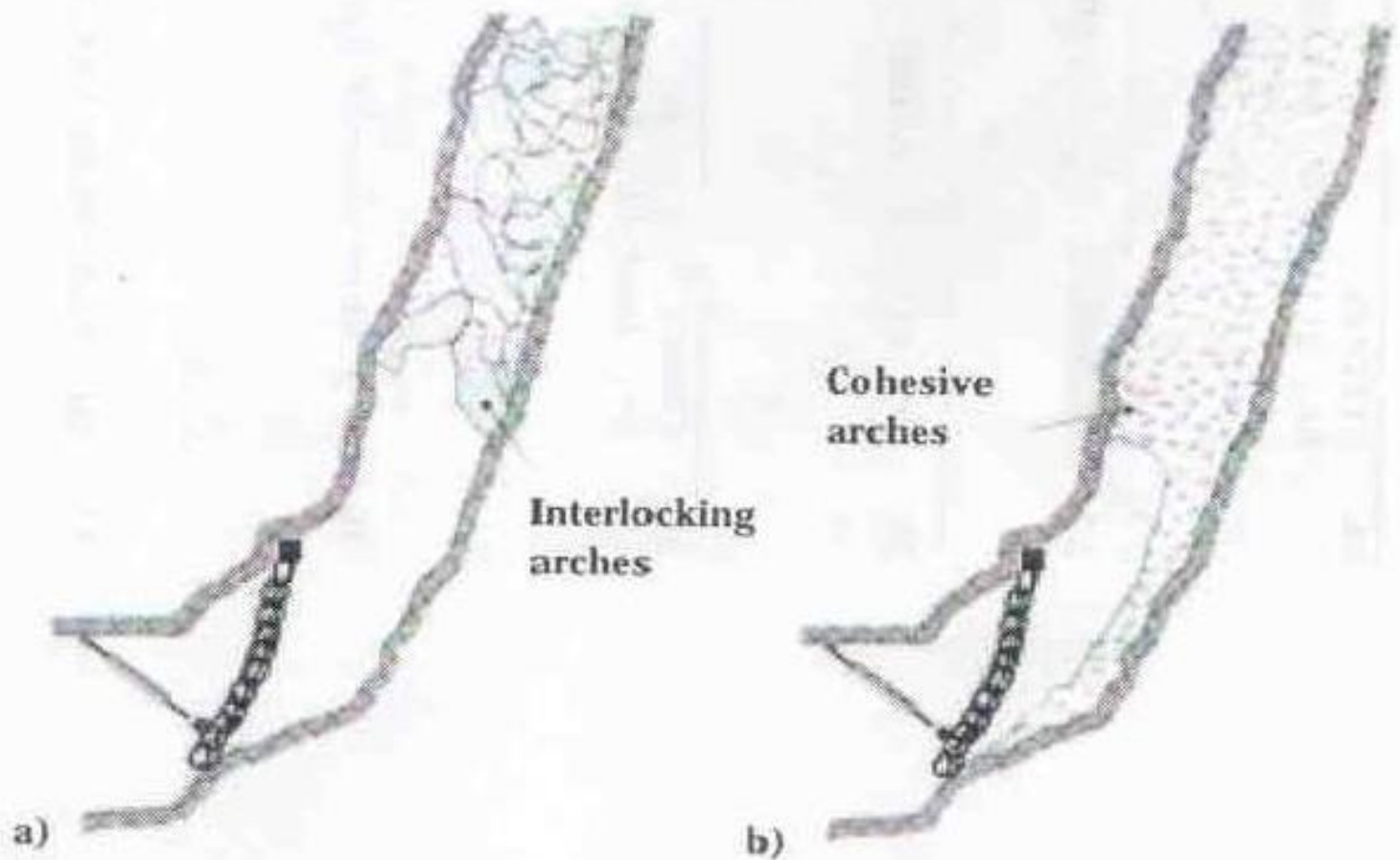
Flow Of Ore IV

- *Course material can hang up in ore pass due to interlocking arches.
- *Transfer of fine material results in hang up due to cohesive arches.
- *Blockage can happen near chute or high up .
- *Hang up in ore pass is more common for mine with paste filling.
- *Hang up can be cleared up by “Sputnik”, an air propelled device, that can carry 12kg explosive 100 high in ore pass.

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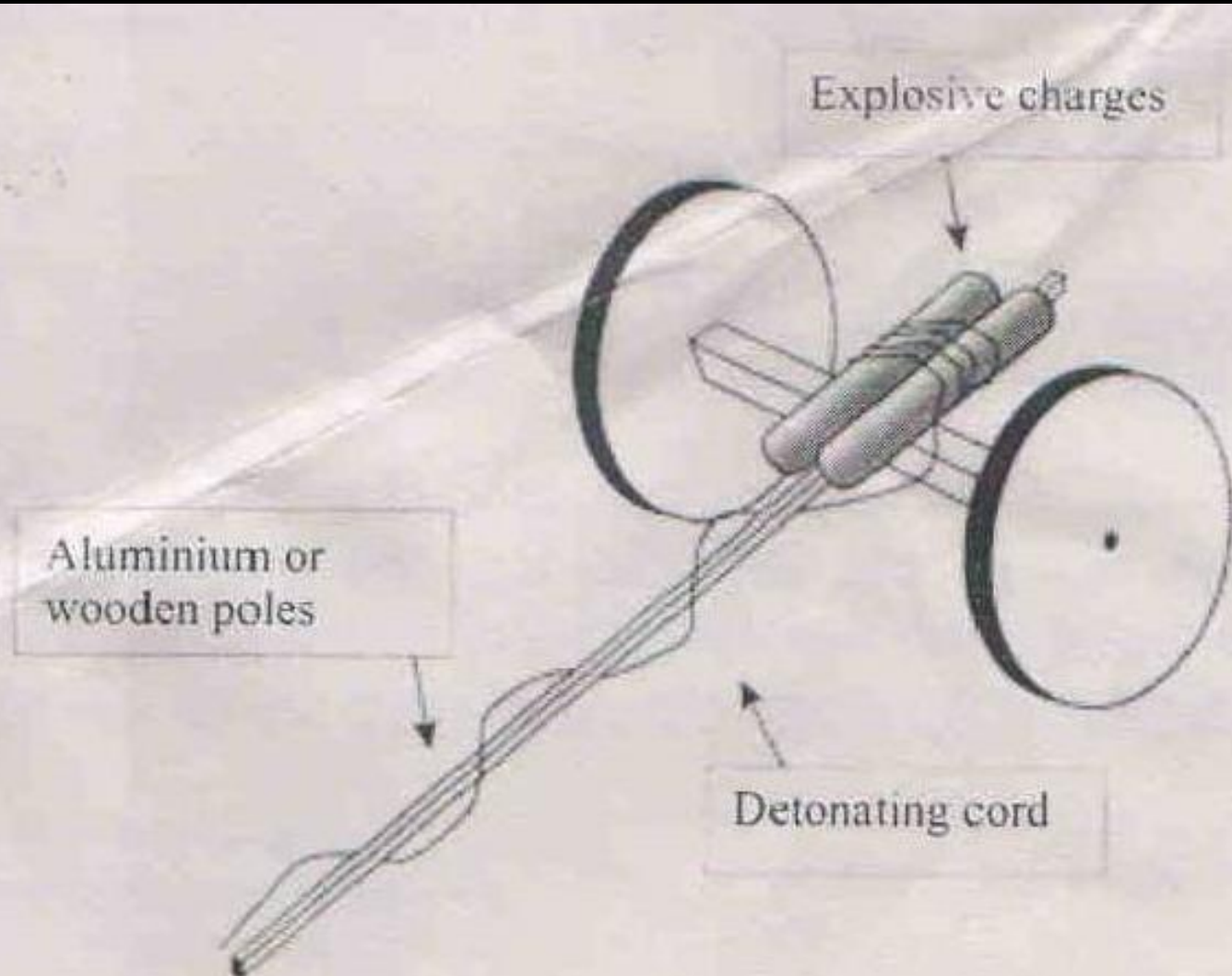
Hangup I



—Hang-ups caused by: (a) interlocking arches (b) cohesive arches

Sputnik I

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-Use of a buggy to drive explosive charges up an ore pass, after Hadjigeorgiou and Lessard (2004)

Sputnik I

Ore pass practice in Canadian mines

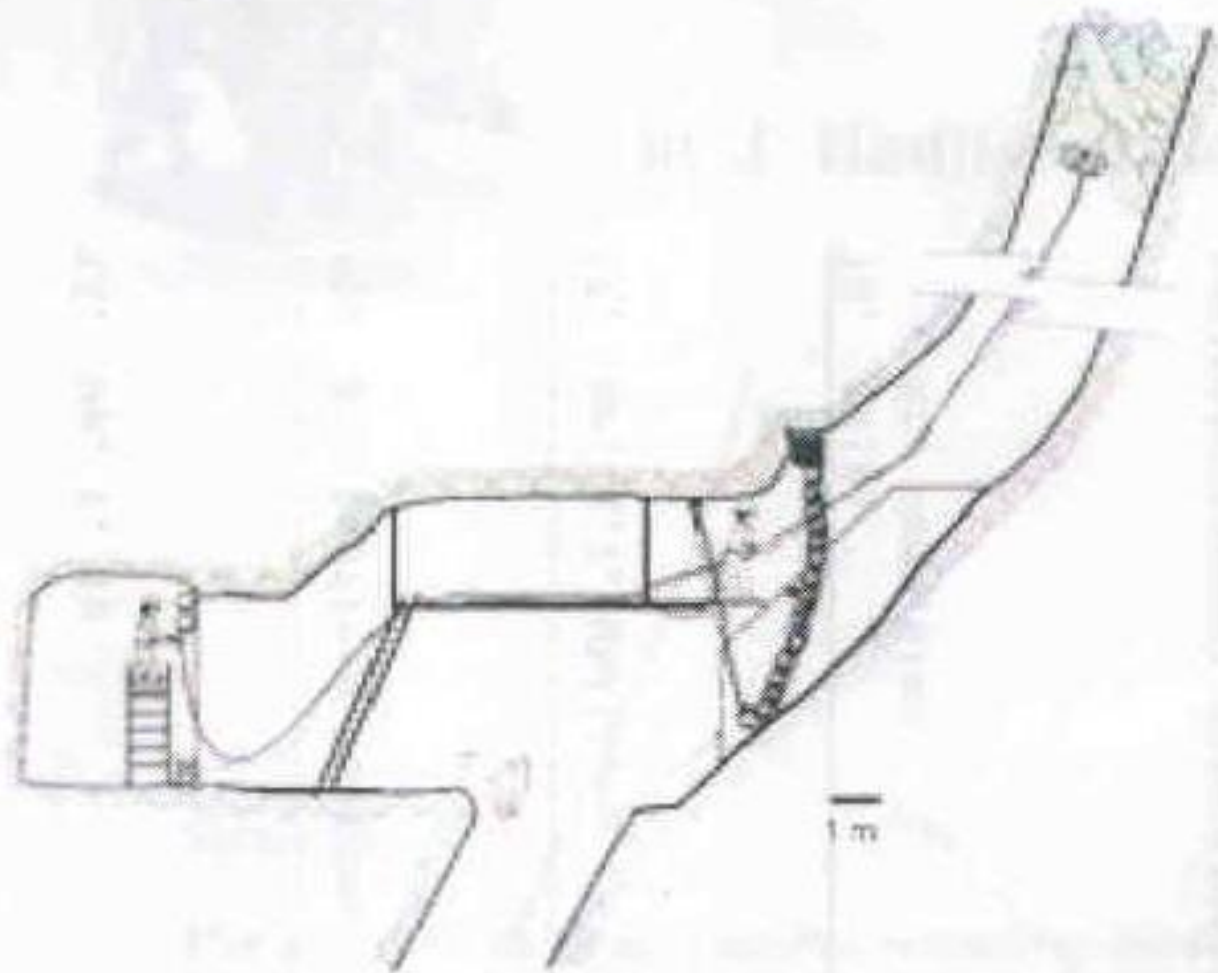


Figure 11—Application of Sputnik to clear an ore pass, after Hadjigeorgiou and Lessard (2004)

Flow Of Ore V

- *Holes can be drilled up to point of hang up & blasted after charging hole with explosive.
- *Use of water to clean hang up should be restricted, to avoid rush of mud flow.

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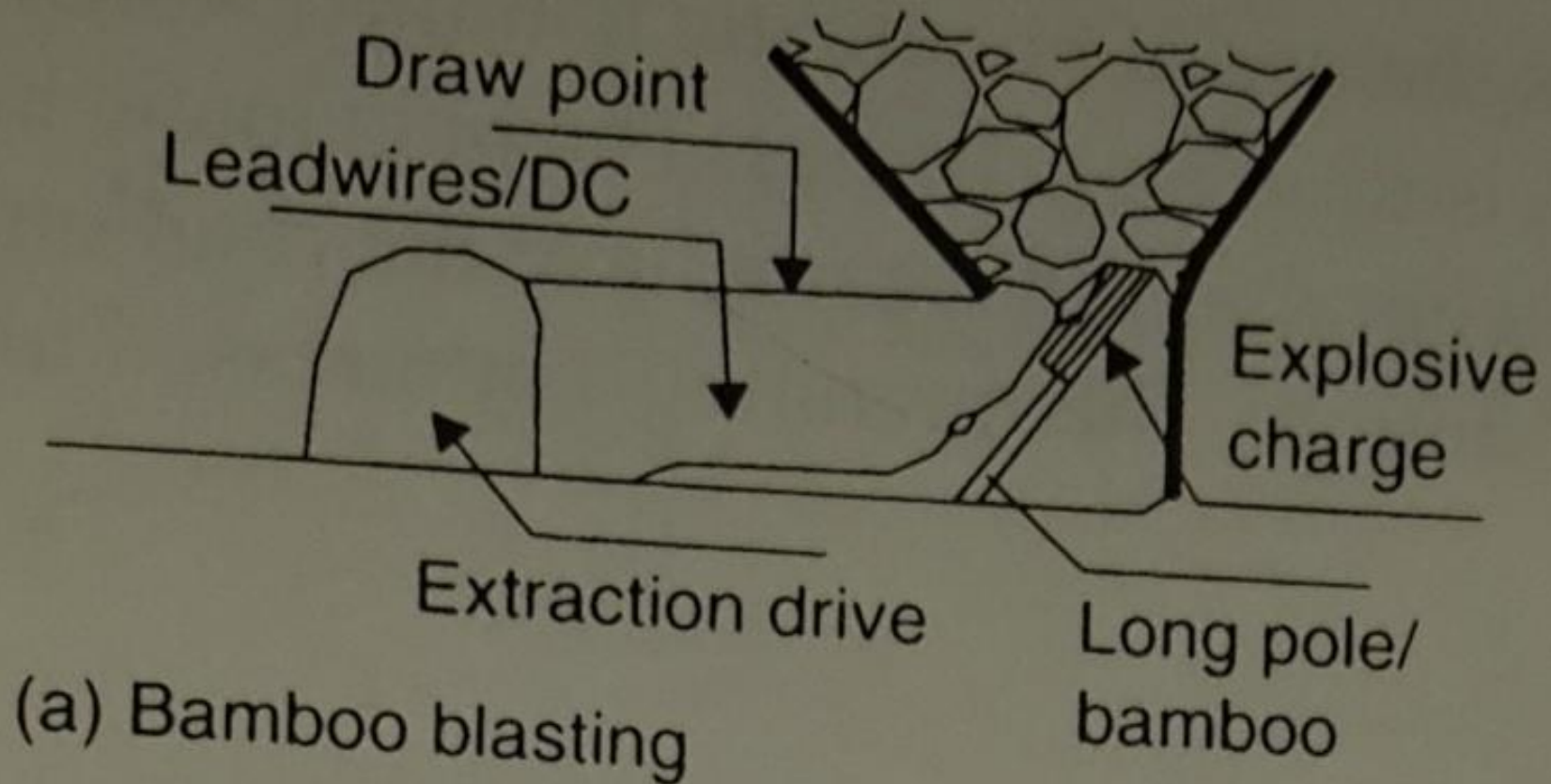
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Ore Pass Bamboo Blasting

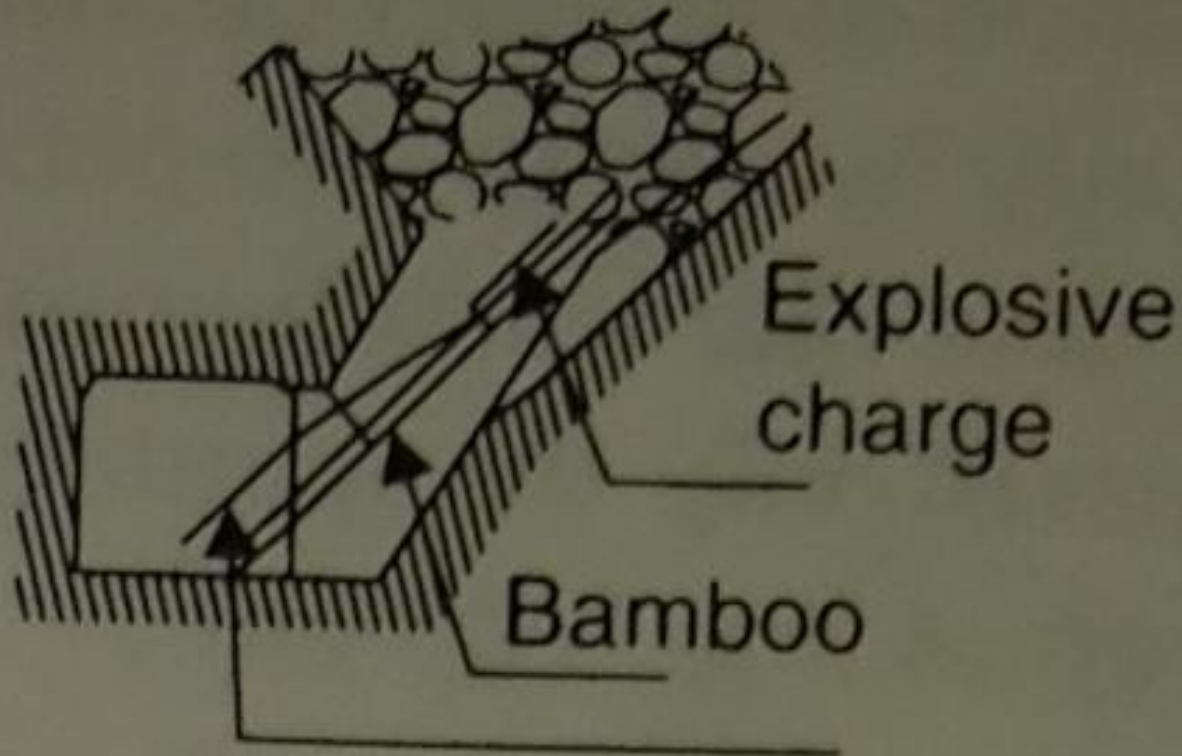
- *Clearing jam at 5 to 6 meter from mouth of ore pass.
Explosive and detonator are tied at end of bamboo.
Manually bamboo is pushed to clogged area and
blasted for removing jam,
- * Bamboo blasting for jam removal.

Ore Pass Bamboo Blasting



Ore Pass Bamboo Blasting

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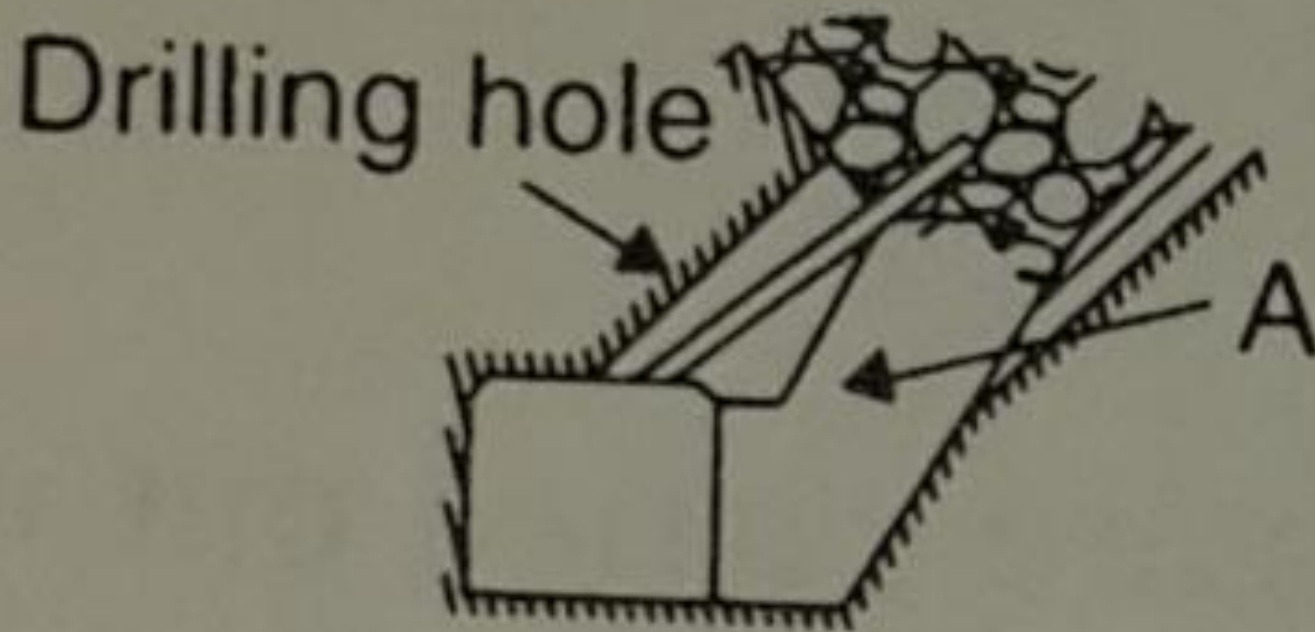
Leadwires/D.C
Bamboo blasting

Ore Pass Shot Drill Hole Blasting I

Short hole is drilled from level near mouth of ore pass. Upward manually drilled hole is of maximum 5 to 6 m.

Ore Pass Shot Drill Hole Blasting

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(b) Drilling a short
hole (SH)

Long Hole Blasting I

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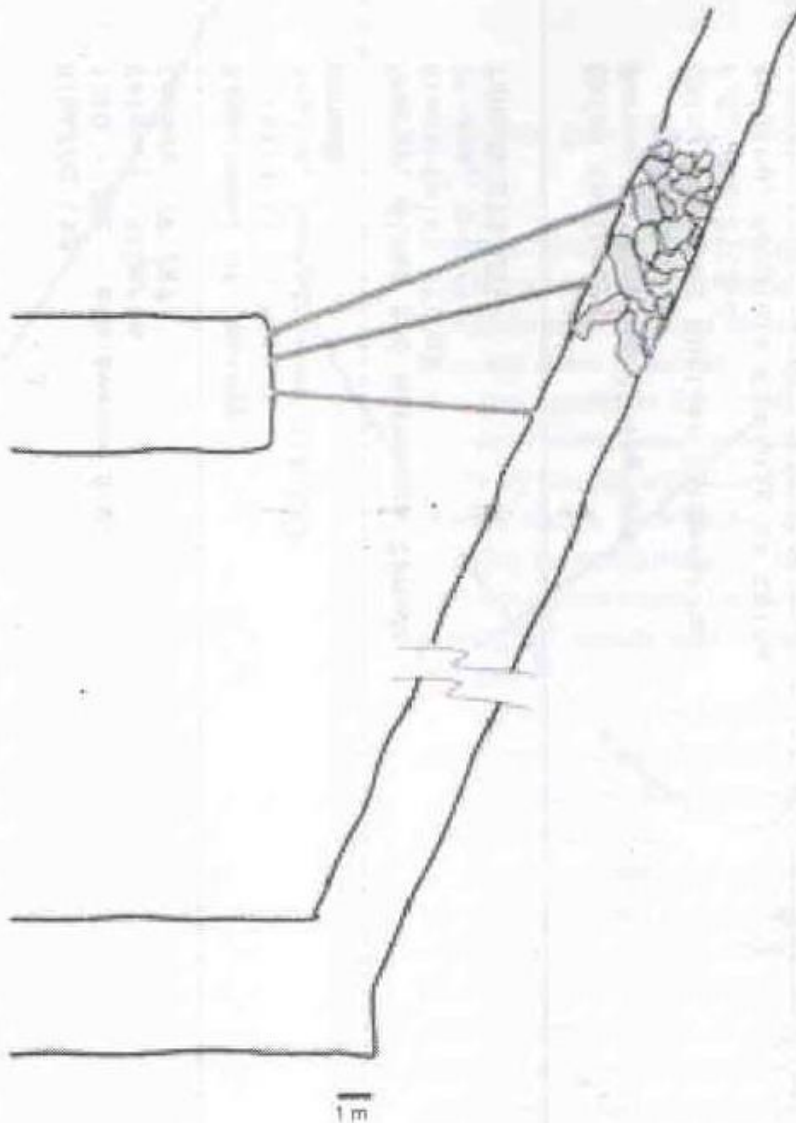


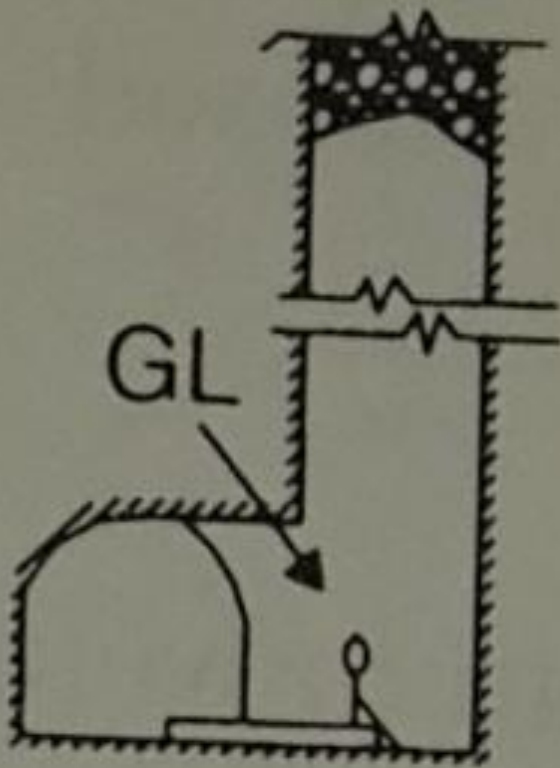
Figure 12—Use of explosives to clear hang-ups high in an ore pass, after HadjiIgeorgiou and Lessard (2004)

Ore Pass Grenade Blasting I

*When, jamming is beyond the range of bamboo or hole, jamming in ore pass is removed with help of Grenade launcher. Grenade explodes while in contact with jammed ore in ore pass.

Ore Pass Grenade Blasting

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(c) Firing a grenade launcher GL

Reinforcement I

- *Supporting ore pass increases life of ore pass.
- *Ore pass excavation by Alimak raise can be supported by rock bolt from working platform.
- *Resin grouted rebar is most common support for ore pass.
- *Resin grouted short cable is becoming popular for ore pass support.
- *Strand Locked coil cable bolt is latest being used.

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Reinforcement II

- *With stand lock-bolt, cable is under tension to achieve pre stressed condition.
- * Rock bolt reinforcement of ore pass creates problem at discharge point by disturbing ore flow.
- *Suitable lining on ore pass surface prevent ore pass expansion & increase life of ore pass.
- *Lined and unlined ore pass.
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Reinforcement III

- *In bad ground, ore passes often lined with concrete. Concrete is faced with high strength steel liner.
- *In hard rock, a long ore pass (glory hole) is not normally lined, instead two ore passes are provided near same location for following reasons.

Reinforcement IV

- *Lining long ore pass is more expansive than drilling another hole.
- *A second raise provided required relief to exhaust air blast (when both raises are inter connected).
- *If one raise becomes inoperable, second one is available.

Monitoring I

- *Condition of ore pass deteriorates with use.
- *Condition is to be monitored to prevent ore pass from collapsing.
- *This also helps to develop 2nd ore pass in time.
- *An ore pass section is considered to have failed, when volume of ore pass has increased to double that of initial ore pass volume.

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Monitoring III

- *Ore pass can be visually inspected from upper level to detect deterioration of wall.
- *Visual inspection method is inaccurate & ore pass has to be emptied & sufficiently illuminated.
- *Information about ore pass is collected by drilling hole at ore pass from near by drift.
- *For drilling, ore pass can be kept full or empty.
- *drilling hole provides only localized data.
- *Ore pass can be monitored by lowering of camera in to ore pass.

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Monitoring IV

- *For monitoring by camera, ore pass has to be properly illuminated.
- *For lowering camera, ore pass has to be empty.
- *Cavity monitoring system (CMS) gives a full 3D profile of ore pass
- *CMS is accurate, with some blind spot.
- *For examination by CMS, ore pass need not be illuminated.
- *CMS is capital intensive method.
- *CMS examination, ore pass should be empty.

Monitoring IV

- * Information available through is in digital form and can be utilized with different software or electronic mode. (modulated infrared energy signal).

Monitoring Muck Level I

- *It is also necessary to evaluate muck level in ore pass.
- *Visual muck level inspection is performed by supervisor from top of ore pass.
- *This system is accurate.
- *Ore level in ore pass is monitored by weighted tape.
- *This tape is lowered by supervisor from top of ore pass periodically to record muck level.
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Monitoring Muck Level II

- *Laser system provides real time muck level reading.
- *This system is very accurate & can be made automated.
- *This system is sensitive to dust.
- *Tonnages reconciliation by subtracting in put & out put from ore pass.
- *Input is measured by number of bucket dumped in ore pass and number of mine car filled at bottom of ore pass.
- *This method is highly inaccurate.

Monitoring Muck Leve III

- *Some mines follow system of keeping ore pass full.
- *This prevents danger to ore pass wall due to impact.
- *Keeping ore pass full require strict monitoring of input & out put of ore pass.

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Reference I

- *Surface and Underground Excavation – Methods, Techniques and Equipment. Ratan Raj Tatiya, Page – 350 to 354.
- *Ore Pass Practice in Canadian Mines. J Hadjigeorgiou, J F Lessender, F Mereieer-Langevin.
- *Ore and Waste Pass in Queens Mines, Arizona USA,